

Bilateral vs. Multilateral Peering at IXPs: A Routing Hygiene-centric Evaluation

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Abstract

Content providers at IXPs are increasingly shifting from multilateral to bilateral peering to gain finer-grained policy control and improve routing reliability. In this paper, we quantify the impact of this design choice by comparing routing outcomes across the two models. We introduce a measurement metric for route quality, called routing hygiene, that captures multiple dimensions of routing correctness and security. Applying our framework for measuring route hygiene to BGP data from multiple IXPs and a large CDN, we find that routing hygiene differs systematically across peering models, with each model exhibiting distinct hygiene profiles and neither uniformly superior across all metrics.

1 Introduction

Internet service providers (ISPs) and content delivery networks (CDNs) have increasingly embraced Internet Exchange Points (IXPs) to localize traffic, reduce transit dependency, and lower network latency, as evidenced by the doubling of the IXPs since 2020 to more than 1,000 active exchanges globally [20]. To scale efficiently, IXPs have implemented multilateral peering through centralized route servers, allowing each participant to peer with every other route server participant using a single BGP session rather than maintain bilateral sessions with every other Autonomous System (AS) connected on the IXP (Figure 1) [14, 37].

Route servers redistribute routes without participating in the data plane, reducing control-plane complexity while preserving dense interconnection among networks according to a centrally configured policy in contrast to per-peer policy definitions in bilateral peering sessions. This centralized policy control can theoretically improve routing hygiene when proper filtering is enforced to restrict propagation of incorrect routing information. In this context, routing hygiene is defined as the adherence to the best BGP routing practices that aim to prevent the propagation of incorrect routing information, such as the MANRS framework [27].

However, when not carefully implemented, multilateral peering can exacerbate the scope of poor routing hygiene, as an insecure route server would broadcast misconfigured or bogus paths to all of its members [12, 37]. Additionally, the configuration simplicity of route servers disappears when IXP members want to implement granular control route redistribution policies, increasing the risk for misconfigured or unintentional policies [16].

Indeed, in recent years, public operator reports indicate that major cloud and content networks are increasingly opting out of route servers to establish bilateral sessions [4, 5, 22, 35]. Such networks prefer bilateral peering which allows networks to fully control which networks they peer with, and to maintain per-peer routing policies, which may reduce the “blast radius” of unhygienic routing. In 2020 PeeringDB introduced the “Never via Route Servers” flag as an attempt to limit the propagation of BGP route leaks through IXP route servers, which has been a persistent pain point until then [41]. Despite the operational importance of routing hygiene over IXP peerings, there is limited attention from the research community.

As a result, there is no consensus in the operational community on whether multilateral peering improves or hinders interdomain routing security [15, 30]. In this paper, we aim to inform this debate with evidence derived from large-scale analysis routing hygiene at IXPs, and present an empirical study examining whether shifting from multilateral to bilateral peering or private network interconnections (PNI) is associated with differences in the observed routing hygiene. We employ two complementary measurement approaches: *Stateless Path Analysis* which detects anomalies observable directly in BGP attributes, e.g., Private ASNs or bogon routes, and *Algorithmic Path Analysis* which requires the post-processing of routing data with external datasets and policy context to detect phenomena like route leaks and RPKI violations.

Our findings reveal that neither peering model is uniformly superior, and there are measurable trade-offs between policy flexibility and routing correctness:

- More frequent stateless hygiene violations in bilateral peering: Bilateral sessions exhibit consistently higher rates of stateless poor hygiene. This includes AS-path prepending (37.1% versus 20.9% in multilateral), AS-path loops (affecting 0.84% versus 0.25% of unique prefixes), and multi-origin AS (MOAS) conflicts, which were absent from multilateral sessions.
- Stronger Algorithmic Validation in multilateral peering: Multilateral sessions achieve higher aggregate validation rates, with greater proportions of IRR-valid (91.25% versus 85.26%) and RPKI-valid (50.13% versus 42.86%) announcements.
- Divergent high-impact security-relevant hygiene violations: Security-relevant hygiene violations manifest

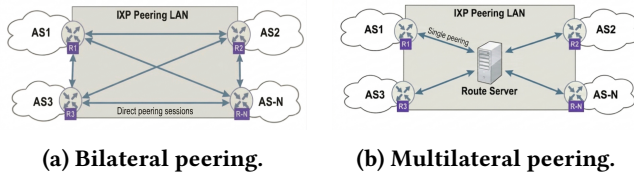


Figure 1: Bilateral peering requires direct, one-to-one BGP sessions between ASes for tight control, while multilateral peering uses a central IXP route server to redistribute routes, making it much easier to scale.

differently across the peering models. The two confirmed route leaks in our dataset were exclusively observed in bilateral sessions, whereas bogon announcements were only found in multilateral sessions.

Code Release: We will make our source code for our analysis framework open-source upon publication.

2 Background

Major content networks have publicly moved toward bilateral peering. Google announced in 2024 it would stop peering via route servers, citing reliability and explicit peer selection [4], AWS requires bilateral peering above a traffic threshold for policy control and traffic engineering predictability [22], and Meta cites filtering scope and “blast radius” containment in its long-tail peering strategy as reasons for avoiding multilateral peering [30]. Smaller operators and peering coordinators echo these concerns [5, 35]. In 2020 PeeringDB’s created the “Never via Route Servers” flag [41] to allow AS operators signal their preference to never have their prefixes announced via route servers.

The rationale across these sources converges on three themes:

- **Per-peer policy expressiveness:** route servers approximate per-peer export policies only through BGP communities, whereas bilateral sessions allow direct configuration of import and export filters on a per-neighbor basis.
- **Reduced blast radius:** a misconfiguration or compromised session at a route server can propagate to every connected member, while bilateral failures are contained to the two sessions involved.
- **predictable traffic engineering:** selective AS-path prepending, MED signaling, and community tagging behave more consistently when applied to a known peer than when redistributed through a centralized policy engine.

The operator community, however, remains divided on whether this shift is net-positive for routing hygiene [15, 30].

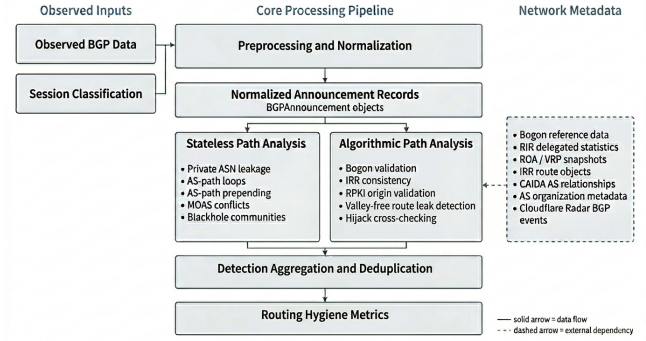


Figure 2: Routing hygiene Analysis Pipeline.

3 Empirical Evaluation

Our approach is to empirically quantify routing hygiene at IXPs by combining observed BGP route data from route collectors and looking glass (LG) servers, with external network metadata to infer a set of routing-hygiene signals. We define routing hygiene as the degree to which BGP announcements observed on a peering session conform to established operational filtering guidelines and routing best practices, such that deviations indicate either policy violations or control-plane behaviors that degrade path selection. Concretely, we operationalize this definition through a set of signals that serve as indicators associated with violations of operational filtering guidelines for IXPs and network operators [27, 28], as well as control-plane behaviors that may negatively impact path selection efficiency [29, 42]. We emphasize that these signals do not uniformly correspond to misconfigurations. While some (e.g., bogons or RPKI-invalid announcements) represent objective policy violations, others (e.g., multi-origin announcements) may arise from legitimate traffic engineering practices. Accordingly, we interpret their prevalence as indicative of routing-hygiene risk rather than definitive evidence of harmful routing behaviour.

3.1 Framework

Figure 2 shows an overview of our framework. It takes as input BGP routes (including AS hops and the next-hop peering interface), and metadata to classify the peering sessions into non-IXP, IXP bilateral or IXP multilateral. For the classification of peering sessions we rely on identifying the IP ranges used by IXPs and their route servers according to PeeringDB, IXP websites and data obtained from the Peering Manager tool¹ deployed by IXPs and a global AS network connected at hundreds of IXPs.

The input paths are then analyzed to extract routing hygiene signals observable in BGP control-plane data. We distinguish between two types of signals: *stateless signals*, which

¹<https://docs.ixpmanager.org/latest/features/peering-manager/>

capture directly observable behaviors from BGP announcement attributes with minimal assumptions (Section 3.1.1), and *algorithmic signals*, which incorporate external metadata to validate context (Section 3.1.2).

3.1.1 Stateless path analysis. We infer the following routing phenomena directly from BGP announcements:

Excessive AS path prepending. AS path prepending is a BGP traffic engineering technique, with which an AS that tries to influence inbound traffic, by repeatedly adding its own AS number to the AS-path of a prefix announcement. This artificially inflates the path length, making the route less preferable to external routing decision engines. *Excessive prepending* occurs when a route contains an anomalously large number of these duplicate ASNs, which is typically the result of misconfiguration rather than a deliberate policy.

Excessive prepending is considered a self-inflicted vulnerability that degrades routing security [25, 29]. Beyond offering diminishing engineering returns, inordinately inflating path lengths makes a network highly susceptible to both accidental route leaks and deliberate prefix interception attacks. Malicious actors can exploit heavily prepended routes via “path stripping”, which involve covertly removing the duplicate ASNs to forge a falsely attractive, shorter path that easily hijacks traffic. This dual vulnerability necessitates a critical re-evaluation of prepending practices, urging operators to eliminate unnecessary prepends to preserve overall routing integrity. The IETF’s Global Routing Operations (GROW) working group recommends for operators to limit the maximum length of the AS paths they accept, although they do not specify specific thresholds [31].

Bogon ASN leakage. Bogon ASNs encompass private, reserved, and unallocated ASNs [19]. Bogon ASNs lack verifiable, legal ownership, so their inclusion in the AS path obfuscates the true origin of route announcements, effectively providing cover for malicious activities such as prefix hijacking and traffic interception. Furthermore, leaking unallocated ASNs introduces latent reachability vulnerabilities when these blocks are subsequently assigned to legitimate networks, residual invalid announcements trigger the loop-prevention mechanisms of BGP, resulting in routing blackholes. Consequently, foundational routing security standards, including RFC7454 (BCP 194) [13] and the MANRS initiative [33], require network operators to implement strict filtering to drop paths containing bogon ASNs.

AS path loops. is a definitive indicator of poor routing hygiene, as it means the failure or deliberate bypass of the inherent loop-prevention mechanism of BGP. By design, BGP will immediately discard any received route update that already contains its own ASN, as defined in the RFC4271 [36]. When AS loops nevertheless appear in the Default-Free Zone

(DFZ), they are overwhelmingly the result of operator misconfiguration, such as the improper use of override commands (e.g. `allowas-in`) to bypass topological constraints. Consequently, enforcing strict path-vector validation and preventing the export of looped paths is a critical cto ensuring that localized misconfigurations do not pollute the global routing system.

AS-path loops identified by removing consecutive prepending-induced duplicates and detecting repeated non-consecutive ASNs,

multi-origin prefixes (MOAS) identified by inconsistent origin ASes for the same prefix, and **blackholing announcements** indicated by the presence of the BGP community 65535:666 (RFC 7999).

3.1.2 Algorithmic path analysis. We infer the following using validation models and complementary network metadata: **bogon announcements** validated against CAIDA and Team Cymru bogon archives after time-aligning datasets (i.e, route collect BGP data with bogon reference dataset)[1], **RPKI-invalid announcements** identified by comparing prefixes and origin ASes against date-aligned ROA payloads (VRPs) from the RIPE NCC RPKI archive under prefix coverage, origin matching, and maxLength constraints, **IRR inconsistencies** detected by comparing announced prefixes against route objects retrieved via bgpq4 for the origin ASN, **route leaks** inferred using the valley-free model augmented with CAIDA AS-relationship and AS-to-organization data to reduce false positives by accounting for sibling and non-standard relationships (e.g., non-commercial and measurement-oriented relationships of networks such as RIPE NCC RIS) , and **route hi-jacks** identified through cross-referencing announcements with Cloudflare Radar BGP hijack events.

3.1.3 Network Size Classification. We classify network size by AS customer degree, using CAIDA AS Rank customer counts and the thresholds introduced in [11]: small for customer degree ≤ 2 , medium for $2 < \text{customer degree} \leq 180$, and large for customer degree > 180 .

4 Datasets

Our analysis combines BGP control-plane data with external metadata to infer routing-hygiene signals across IXP peering models. To ensure temporal consistency, we retrieve historical snapshots of all metadata datasets (e.g., CAIDA, RIPE NCC) aligned to the timestamp of each analyzed BGP snapshot. This avoids mismatches between routing observations and validation context, which is critical for time-dependent signals such as bogon classification and RPKI validation. Additionally, we worked with a large CDN to capture data to demonstrate that our trends holds (Section 4.1).

4.1 CDN Dataset

we leverage BGP data from a large global CDN spanning over 500 Points of Presence (PoPs). We obtain Routing Information Base (RIB) snapshots from the CDN’s BMP-based analytics pipeline for a single day in April 2026, capturing BGP state across all edge routers. We classify each BGP session using the CDN’s peering metadata as: (i) *IXP multilateral* (via route servers), (ii) *IXP bilateral* (direct peering at an IXP fabric), or (iii) *non-IXP* (private interconnect or transit). The dataset includes over 60,000 BGP sessions across 348 IXPs, including approximately 51,600 IXP sessions (of which 1,347 are route-server sessions)², and approximately 8,400 non-IXP sessions. This dataset enables a global, operational view of routing behavior across peering types.

4.2 BGP Data from IXPs

To complement the large scale CDN commercial, we analyze BGP routing data collected from Packet Clearing House (PCH) route collector snapshots [34] at three IXPs: DE-CIX Frankfurt, DE-CIX New York, and NAPAfrica Johannesburg. For DE-CIX Frankfurt, we use 14 snapshots spanning April 2019 to August 2025 (collected biannually in April and August). For DE-CIX New York, we analyze 8 snapshots from April 2019 to August 2022. For NAPAfrica Johannesburg, we include a snapshot collected on 24 October 2025. In total, our dataset comprises 23 MRT snapshots covering both bilateral and route-server-mediated multilateral peering sessions.

4.3 Metadata for Validation

We use multiple external datasets to provide validation context for algorithmic analysis.

CAIDA datasets. We use CAIDA’s AS Relationships dataset [7] (serial-2) to infer provider–customer and peer-to-peer relationships for valley-free route-leak detection. We complement this with the CAIDA AS-to-Organization mapping [9] to identify sibling AS relationships and reduce false positives. We also use CAIDA AS Rank [6] to classify network size.

RPKI data. We perform origin validation using Route Origin Authorization (ROA) data from the RIPE NCC RPKI archive [38], covering all five Trust Anchor Locators (TALs). These datasets enable classification of announcements as RPKI-valid, invalid, or unknown.

IRR data. IRR consistency is assessed by retrieving route objects for origin ASNs via bgpq4 [3] and comparing announced prefixes against registered entries.

Bogon data. We detect bogon announcements using CAIDA’s CYMRU bogon archive [8] and Team Cymru’s fullbogon dataset [40]. These datasets include both traditional bogons and unallocated address space, and are aligned temporally

²Multiple route servers per IXP and multiple ASNs per CDN contribute to this count

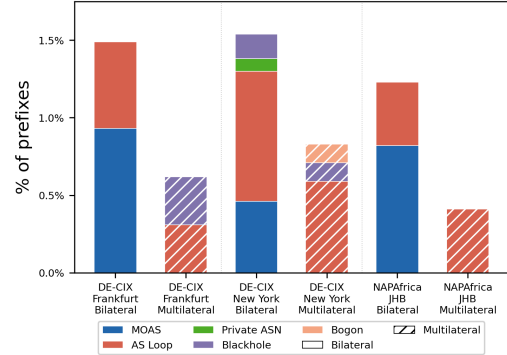


Figure 3: Analysis Stateless anomalies.

with each BGP snapshot to account for time-varying allocations.

Hijack data. We use the Cloudflare Radar API [10] to identify known BGP hijack events for cross-validation of detected anomalies.

5 Preliminary Results

5.1 Stateless Path Analysis

Our analysis reveals that bilateral sessions higher stateless anomaly rates compared to multilateral sessions at the three analysed IXPs, as shown in Figure 3. We discuss each anomaly type below.

AS path prepending. AS-path prepending is the most prevalent stateless anomaly, appearing more often in bilateral sessions (37.1%) than in multilateral sessions (20.9%). Multilateral sessions are more concentrated at no prepend and a single prepend, whereas bilateral sessions are more represented at 2–3 prepends, with a smaller difference at 4 or more prepends showing that prepending is more common in bilateral sessions. We also find that smaller networks are more likely to prepend than larger ones. At IXPs, prepending primarily supports traffic engineering, but it can be harmful by making competing bogus routes more attractive during leaks or hijacks [29]. The highest prepend depth of 27 is observed in bilateral followed by a depth 25 under multilateral, current industry best-practice guide to use AS-path prepending selectively and avoid excessive prepending[?].

Multi-origin AS (MOAS). We observed intra-organization MOAS in bilateral sessions (0.87%), these are representative of traffic engineering by multi-AS organisations like Amazon. Despite being operationally legitimate, they remain visible to external peers as origin conflicts and are prone to being filtered out.

AS path loops. AS-path loops were more prevalent in bilateral than in multilateral sessions, affecting 0.84% of bilateral peer-prefix pairs and 0.25% of multilateral pairs across five

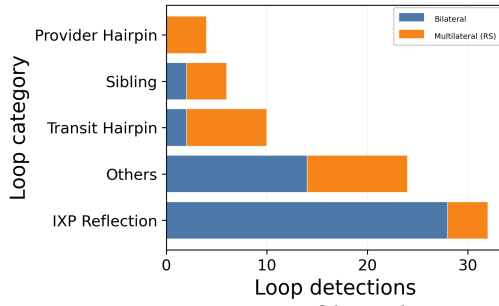


Figure 4: Categorisation of loop detections

structural categories. As shown in Figure 4, we further categorise the observed loops and find that bilateral cases are dominated by *IXP route-server reflection*, in which a participant that engages in both bilateral and route-server peering re-announces routes learned via the route server into its bilateral sessions, causing its own ASN to reappear in the path. In contrast, multilateral cases are more often associated with provider and transit hairpins. *Provider hairpins* arise when a multihomed customer connects to the same upstream provider at multiple attachment points, causing the customer ASN to reappear after traversal of that provider’s backbone. *Transit hairpins* similarly involve ASN reappearance, but after traversal through another transit AS rather than through multiple attachment points of the same provider. *Sibling and merger* cases are observed in both session types, with the remaining detections grouped as *Others*. Although some loops may reflect intentional traffic engineering, prior work shows that a fraction correlate with forwarding loops or are consistent with misconfiguration and policy violations [42].

Blackhole community. Three blackhole-tagged peer-prefix pairs were observed, all announcing /32 host prefixes. Two were observed in bilateral sessions and one in a multilateral session. Although blackholing is a legitimate mitigation mechanism, propagation beyond the intended scope can cause collateral reachability loss or outages.

Private ASN leakage. A single bilateral session at DE-CIX New York was observed announcing a path containing private ASN 65001 (RFC 6996), attributed to AS267613. The presence of a private ASN in a globally visible AS path indicates incomplete eBGP boundary filtering at the originating network. No such leakage was observed in multilateral sessions, consistent with route server AS path sanitization removing private ASNs from redistributed announcements as a standard filtering step.

5.2 Algorithmic Analysis

Algorithmic analysis reveals anomalies under both peering models. However, the highest-impact events in our dataset, particularly route leaks, were observed only in bilateral sessions.

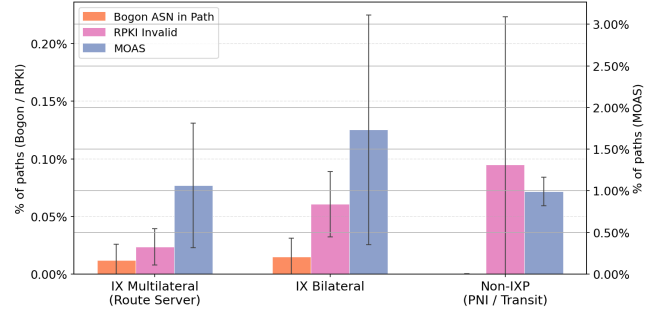


Figure 5: CDN’s BGP paths exhibiting different route hygienen properties: IXP Multilateral (route server), IXP Bilateral, and Non-IXP (PNI / Transit)

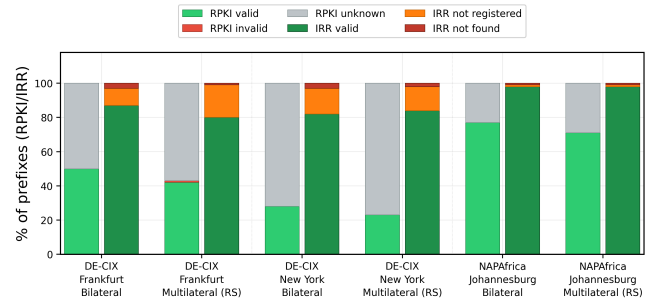


Figure 6: IRR and RPKI filtering.

Bogon prefixes. We found two duplicate bogon prefix announcements of the prefix 2001:df1:ad80::/47 through a multilateral peering session. Taking into account the time-dependent nature of bogon detections, this particular prefix was located to APNIC but not assigned by APNIC within the observation period. The propagation of such prefixes is a routing-hygiene concern because it indicates that boundary filtering did not fully prevent their announcement and may increase operational overhead, as reported in [1].

IRR consistency. As shown in Figure 6, IRR filtering is consistently high across all IXPs and peering models, with multilateral sessions exhibiting a higher proportion of IRR-valid announcements (91.25%) than bilateral sessions (85.26%). The anomalous cases are dominated by announcements classified as “Not Registered” (8.55%), representing prefixes not registered to their originating ASNs, while 2.33% are classified as “Not Found,” indicating that the originating ASNs are not associated with any IRR registry. These cases include observations involving AS269575, AS270559, AS262900, and AS269362. These anomalies are consistent with known limitations of IRR data, including stale, incomplete, or inconsistently maintained route objects [21].

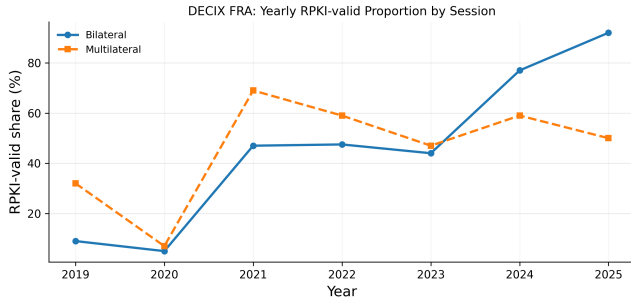


Figure 7: RPKI-valid prefixes over time

RPKI validation. We find that 46.00% of announcements are RPKI-valid, lower than the proportion of IRR-valid announcements, indicating limited RPKI coverage. Multilateral sessions have a higher share of RPKI-valid announcements than bilateral sessions (50.13% vs. 42.86%) and a lower share of Unknown announcements as shown in Figure 6. We also observe a single RPKI-invalid prefix through a multilateral session. Unknown announcements remain the largest category overall (53.99%), indicating that a substantial fraction of routes still falls outside current ROA coverage. Our analysis further shows that large networks are more likely to authorize their prefixes than medium-sized and small networks.

Route leaks and hijacks. Two route leak incidents were identified in bilateral sessions, representing clear violations of the Gao–Rexford model. The first involves a regional provider incorrectly exporting customer-learned prefixes to a peer, while the second shows a content network providing unauthorized transit between its upstream provider and a peer, in violation of valley-free routing principles. No hijacks were observed during the analysis window.

5.3 Longitudinal

We analyse longitudinal trends at DE-CIX Frankfurt using biannual snapshots collected between April 2019 and August 2025. Bilateral sessions account for the majority of observed peer-prefix pairs throughout the observation window. On the stateless side, prepending provides the clearest signal for longitudinal analysis, as the remaining anomaly classes occur too infrequently to support meaningful trend assessment. Bilateral sessions exhibit higher prepending rates from 2022 onward, consistent with greater use of bilateral peering for traffic engineering in the later years. On the algorithmic side, high-impact anomalies such as route leaks, hijacks, and bogon announcements remain sparse and do not exhibit a detectable longitudinal trend under either model. As shown in Figure 7, bilateral sessions show strong growth in RPKI-valid proportions over the observation period and exceed multilateral sessions in the most recent observations, whereas multilateral sessions peak earlier and then fluctuate at lower

levels. Together, these trends indicate that bilateral sessions exhibit increasing stateless anomaly rates driven by prepending, while also showing stronger recent improvement in RPKI-valid proportions than multilateral sessions.

5.4 Analysis of From CDN Perspective

For each prefix-origin pair observed in the RIB, we perform four route quality assessments: (i) bogon detection, validating prefixes against the Team Cymru fullbogon reference lists comprising approximately 159,000 IPv4 and IPv6 unallocated or reserved prefixes, and checking AS paths for reserved or private-use ASNs per IANA registries; (ii) RPKI origin validation per RFC6811 [32], validating each prefix-origin pair against approximately 840,000 Validated ROA Payloads (VRPs) obtained from RIPE’s RPKI validator; and (iii) Multiple Origin AS (MOAS) conflict detection, identifying prefixes originated by more than one autonomous system. To ensure a fair comparison across peering types, we restrict our analysis to common prefixes that are reachable via both IXP and non-IXP paths to eliminate biases from different prefix coverage between peering types.

Note that MOAS conflicts are observed at significantly higher rates (1–7% of prefixes depending on the IXP), because the majority of MOAS cases often reflect legitimate operational practices rather than routing anomalies. Common benign causes include anycast deployments where the same prefix is originated from multiple ASNs at different locations, multi-homed organisations using separate ASNs for different regions or business units, DDoS mitigation, and transit providers re-originating customer prefixes. Unlike bogon or RPKI-invalid routes, MOAS conflicts are not subject to automated filtering, as distinguishing malicious hijacks from legitimate multi-origin announcements requires additional context such as AS relationship data or organisational records. Consequently, MOAS serves as an observational indicator of routing complexity rather than a definitive measure of route hygiene, and its higher prevalence should not be interpreted as evidence of weaker filtering.

6 Related work

Prior work has examined routing hygiene from multiple perspectives, including data-plane measurements of bogon and private-prefix exposure [1], active probing of source address validation [24], and evaluations of MANRS conformance via RPKI and IRR deployment [12], highlighting persistent gaps in filtering and validation. Other efforts study IRR–RPKI inconsistencies [21] or develop techniques to detect route leaks and hijacks using large-scale routing data [18, 23], while IXP studies reveal structural differences between multilateral and bilateral peering [2].

Work specifically on routing hygiene at IXPs is limited. Servillo et al. [39] show that strict route-server filtering is critical to prevent propagation of invalid routes. Madory et al. [26] find inconsistencies in IXP filtering practices, with most dropped routes due to IRR discrepancies rather than RPKI violations. Hlavacek et al. [17] demonstrate that RPKI filtering at route servers provides only localized protection.

In contrast, our work develops a unified measurement framework to compare routing hygiene across bilateral and multilateral peering at IXPs, enabling direct empirical evaluation of these operational models.

7 Discussion and Limitations

This work is a first step that offers a broad characterization of routing hygiene across peering models rather than an exhaustive account. We highlight limitations below:

Our framework relies on passive BGP control-plane data augmented with external metadata, and is subject to three limitations. First, route-collector visibility is partial, missing filtered or suppressed announcements. Second, our inferences depend on the coverage and accuracy of underlying datasets, particularly AS-relationship data for route-leak detection. Third, we rely on heuristics to infer sibling and non-standard AS relationships, which may be incomplete. Our analysis also excludes hygiene dimensions not reliably inferable from control-plane data alone, e.g., source address validation, ASPA-based path authorization, and BGP community misuse, which require data-plane measurements or operator-specific semantics beyond our current scope.

8 Conclusion

We asked whether CDNs are right to prefer bilateral over multilateral peering at IXPs. Based on BGP data from three IXPs and over 60,000 sessions at a large global CDN, the answer is: it depends on what you optimize for. Bilateral sessions exhibit higher rates of prepending, path loops, MOAS conflicts, and the only two confirmed route leaks in our dataset. Multilateral sessions, sanitized by route-server filtering, achieve higher IRR- and RPKI-valid proportions, yet still propagate the only bogons we observe. At DE-CIX Frankfurt, bilateral sessions show the strongest recent gains in RPKI validity, narrowing a gap that once favored route servers. The industry shift toward bilateral peering is thus neither vindicated nor condemned by routing hygiene alone: route servers remain the stronger default for networks that rely on centralized filtering; bilateral peering rewards operators disciplined enough to enforce their own.

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